LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 5
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1. TITLE

Develop a Product Calculator Using HTML, CSS and JavaScript

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2. AIM

To develop an interactive Product Calculator using HTML, CSS and JavaScript that accepts product name, price and quantity, dynamically calculates the total price, and displays the updated product information on the web page.

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3. OBJECTIVE

The objectives of this experiment are:
● To understand the basic structure of an HTML web page.
● To design a user-friendly interface using CSS.
● To understand the use of JavaScript for dynamic web page manipulation.
● To access HTML elements using the Document Object Model (DOM).
● To handle user input dynamically using JavaScript event listeners.
● To calculate the total price based on price and quantity.
● To implement increment and decrement buttons for quantity.
● To perform basic input validation.
● To dynamically update product details and calculation results without reloading the page.

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4. THEORY

HTML (HyperText Markup Language) is used to create the basic structure of the web page. In this application, HTML is used to create input fields for product name, price and quantity along with buttons and display areas.

CSS (Cascading Style Sheets) is used to design and style the Product Calculator. It controls the layout, colors, spacing, borders, buttons, input fields and responsive behavior of the application.

JavaScript is used to provide dynamic functionality to the Product Calculator. It reads the values entered by the user, performs the calculation and updates the web page automatically.

The Document Object Model (DOM) represents the HTML document as objects that can be accessed and modified using JavaScript. In this application, document.getElementById() is used to access HTML elements.

The main operations performed by JavaScript are:

● Reading the product name, price and quantity from input fields.
● Converting input values into numbers using Number().
● Validating price and quantity values.
● Calculating the total using:

Total Price = Price × Quantity

● Updating the displayed price, quantity and total dynamically.
● Displaying the selected product name.
● Increasing and decreasing quantity using the + and − buttons.
● Using event listeners to respond immediately to user actions.

The application uses the following JavaScript events:

● input event – updates the calculation whenever the user changes the input.
● click event – handles the quantity increment and decrement buttons.

The basic flow of the application is:

User Input → JavaScript Event → DOM Processing → Calculation → Dynamic Page Update

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5. ALGORITHM / PROCEDURE

1. Create a new HTML file for the Product Calculator application.
2. Define the basic HTML document structure using `html`, `head` and `body` elements.
3. Add a heading and description for the Product Calculator.
4. Create an input field to accept the product name.
5. Create an input field to accept the product price.

6. Create a quantity input field with a default value of 1.
7. Add + and – buttons to increase or decrease the quantity.
8. Create a summary section to display price, quantity and total price.
9. Use CSS to style the calculator interface and make it responsive.
10. Use JavaScript to access the required HTML elements using `document.getElementById()`.
11. Create a `calculateTotal()` function to perform the calculation.
12. Convert the price and quantity values into numbers using `Number()`.
13. Validate the price and quantity values.
14. Set the price to 0 if an invalid or negative price is entered.
15. Set the quantity to 1 if an invalid value or a value less than 1 is entered.
16. Calculate the total price using **Price × Quantity**.
17. Display the formatted price, quantity and total on the web page.
18. Display the product name dynamically when the user enters it.
19. Add `input` event listeners to the product name, price and quantity fields.
20. Add `click` event listeners to the + and – buttons.
21. Prevent the quantity from decreasing below 1.
22. Call the `calculateTotal()` function initially to display the default values.
23. Execute the application in a web browser.
24. Perform multiple test cases and compare the expected and actual results.

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6. PROGRAM

Code.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Product Calculator</title>

    <style>
        * {
            box-sizing: border-box;
        }

        body {
```

```
    margin: 0;
    font-family: Arial, sans-serif;
    background: #f3eee7;
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
}

.container {
    width: 420px;
    background: #fffdf9;
    padding: 30px;
    border-radius: 14px;
    border: 1px solid #e5ddd2;
    box-shadow: 0 8px 25px rgba(70, 55, 40, 0.15);
}

h1 {
    text-align: center;
    margin: 0 0 8px;
    color: #3d3027;
}

.subtitle {
    text-align: center;
    color: #8a7b70;
    margin-bottom: 25px;
    font-size: 14px;
}

label {
    display: block;
    margin: 15px 0 7px;
    font-weight: bold;
    color: #4b3b31;
}

input {
```

```
    width: 100%;
    padding: 12px;
    border: 1px solid #d6c9bc;
    border-radius: 7px;
    font-size: 15px;
    outline: none;
    background: white;
}

input:focus {
    border-color: #c56a35;
    box-shadow: 0 0 0 3px rgba(197, 106, 53, 0.12);
}

.quantity-box {
    display: flex;
    gap: 8px;
}

.quantity-box input {
    text-align: center;
}

.qty-btn {
    width: 45px;
    border: none;
    border-radius: 7px;
    background: #c56a35;
    color: white;
    font-size: 20px;
    cursor: pointer;
}

.qty-btn:hover {
    background: #a95427;
}

.summary {
    margin-top: 25px;
```

```
            padding: 18px;
            background: #f8f2eb;
            border-radius: 10px;
        }

        .row {
            display: flex;
            justify-content: space-between;
            margin: 10px 0;
            color: #6d5d50;
        }

        .total {
            border-top: 1px solid #dfd2c5;
            padding-top: 12px;
            margin-top: 12px;
            font-size: 22px;
            font-weight: bold;
            color: #a95427;
        }

        .product-display {
            text-align: center;
            margin-top: 18px;
            color: #806f62;
            font-size: 14px;
        }

        @media (max-width: 500px) {
            .container {
                width: 90%;
                padding: 25px;
            }
        }
    </style>
</head>

<body>
```

```
<div class="container">

    <h1>Product Calculator</h1>

    <p class="subtitle">
        Enter the details to calculate the total
    </p>

    <label for="productName">Product Name</label>

    <input
        type="text"
        id="productName"
        placeholder="Example: Wireless Mouse"
    >

    <label for="price">Price</label>

    <input
        type="number"
        id="price"
        placeholder="Enter price"
        min="0"
    >

    <label>Quantity</label>

    <div class="quantity-box">

        <button class="qty-btn" id="minus">−</button>

        <input
            type="number"
            id="quantity"
            value="1"
            min="1"
        >

        <button class="qty-btn" id="plus">+</button>
```

```
    </div>

    <div class="summary">

        <div class="row">
            <span>Price</span>
            <span id="priceDisplay">₹0.00</span>
        </div>

        <div class="row">
            <span>Quantity</span>
            <span id="quantityDisplay">1</span>
        </div>

        <div class="row total">
            <span>Total</span>
            <span id="total">₹0.00</span>
        </div>

    </div>

    <div class="product-display" id="productDisplay">
        No product selected
    </div>

</div>

<script>

    // Getting elements from the page

    const productName =
        document.getElementById("productName");

    const price =
        document.getElementById("price");

    const quantity =
```

```
    document.getElementById("quantity");

const priceDisplay =
    document.getElementById("priceDisplay");

const quantityDisplay =
    document.getElementById("quantityDisplay");

const total =
    document.getElementById("total");

const productDisplay =
    document.getElementById("productDisplay");

const plus =
    document.getElementById("plus");

const minus =
    document.getElementById("minus");


// Function to calculate the total price

function calculateTotal() {

    let priceValue = Number(price.value);

    let quantityValue = Number(quantity.value);


    // Prevent invalid values

    if (isNaN(priceValue) || priceValue < 0) {
        priceValue = 0;
    }

    if (isNaN(quantityValue) || quantityValue < 1) {
        quantityValue = 1;
        quantity.value = 1;
```

```
    }


    // Calculate total

    let totalPrice =
        priceValue * quantityValue;


    // Update the page

    priceDisplay.innerText =
        "₹" + priceValue.toFixed(2);

    quantityDisplay.innerText =
        quantityValue;

    total.innerText =
        "₹" + totalPrice.toFixed(2);


    // Display product name

    if (productName.value.trim() !== "") {

        productDisplay.innerText =
            "You are buying: " +
            productName.value;

    } else {

        productDisplay.innerText =
            "No product selected";
    }
}


// Update total when price changes
```

```
price.addEventListener(
    "input",
    calculateTotal
);


// Update total when quantity changes

quantity.addEventListener(
    "input",
    calculateTotal
);


// Update product name display

productName.addEventListener(
    "input",
    calculateTotal
);


// Increase quantity

plus.addEventListener(
    "click",
    function () {

        quantity.value =
            Number(quantity.value) + 1;

        calculateTotal();
    }
);


// Decrease quantity

minus.addEventListener(
```

```
        "click",
        function () {

            if (Number(quantity.value) > 1) {

                quantity.value =
                    Number(quantity.value) - 1;

                calculateTotal();
            }
        }
    );


    // Calculate initial value

    calculateTotal();

  </script>

</body>
</html>
```

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7. CODE EXPLANATION

Code / Function	Explanation
`<!DOCTYPE html>`	Defines the document as an HTML5 document.
`<meta name="viewport">`	Makes the web page responsive on different screen sizes.
`.container`	Creates the main Product Calculator interface.

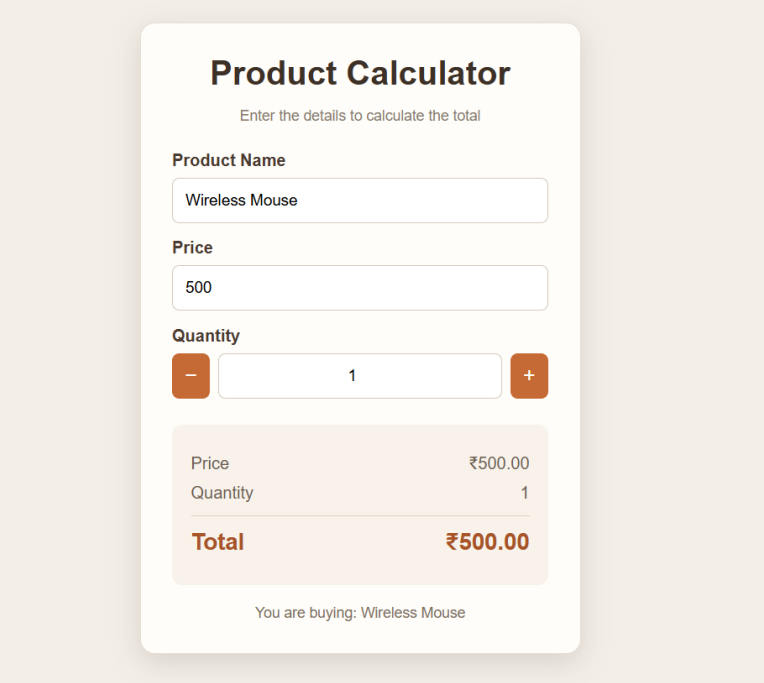
`input`	Accepts product name, price and quantity from the user.
`document.getElementById()`	Accesses HTML elements through the DOM.
`Number()`	Converts input values into numerical values.
`isNaN()`	Checks whether the entered value is a valid number.
`calculateTotal()`	Calculates and updates the product total.
`priceValue * quantityValue`	Calculates the total product price.
`.toFixed(2)`	Displays the amount with two decimal places.
`innerText`	Dynamically changes the content displayed on the page.
`addEventListener()`	Executes a function when a specified event occurs.
`"input"`	Detects changes made in input fields.
`"click"`	Detects clicks on the quantity buttons.
`plus` button	Increases the quantity by 1.
`minus` button	Decreases the quantity by 1, but not below 1.
`trim()`	Removes leading and trailing spaces from the product name.
`calculateTotal()` at the end	Displays the initial values when the page loads.

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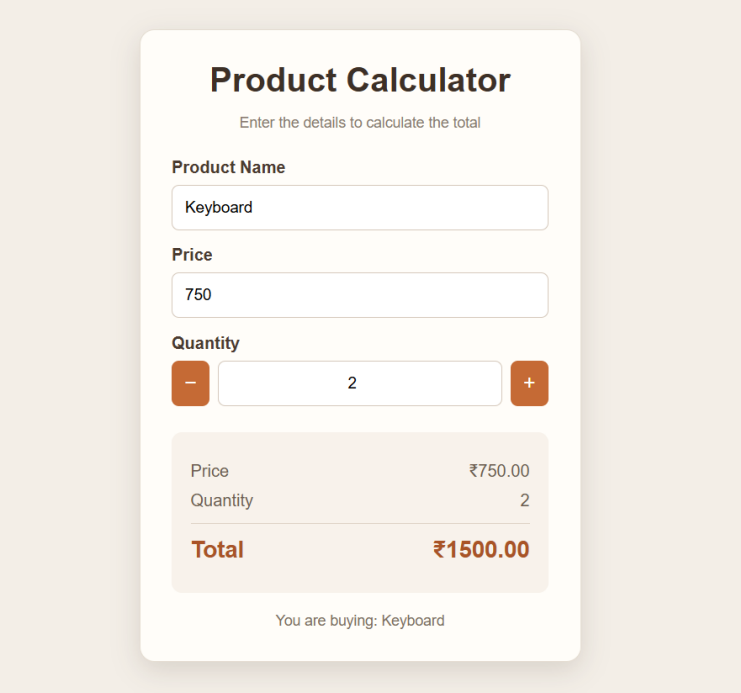
8. TEST CASES AND EXECUTION DATA

Test Case	Product Name	Price	Quantity	Expected Result
TC-01	Wireless Mouse	500	1	Total should be ₹500.00
TC-02	Keyboard	750	2	Total should be ₹1500.00
TC-03	Headphones	1200	3	Total should be ₹3600.00
TC-04	USB Cable	250	5	Total should be ₹1250.00

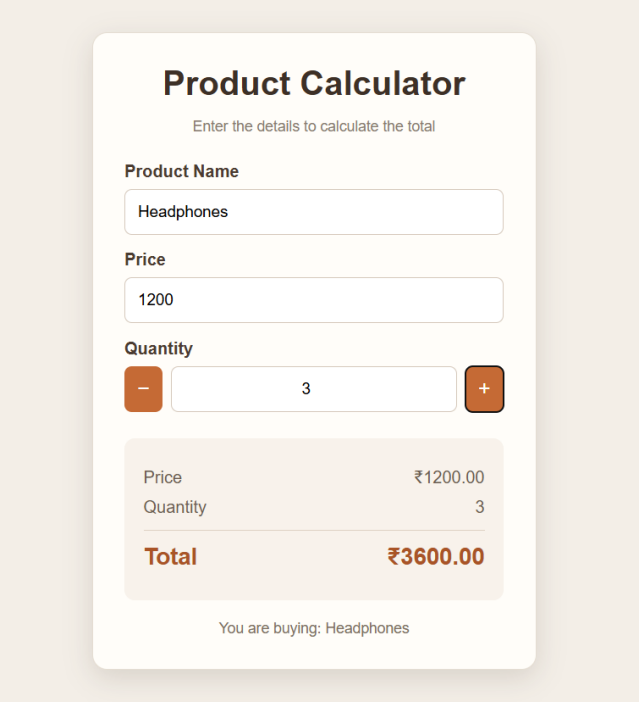
TC-01 output:



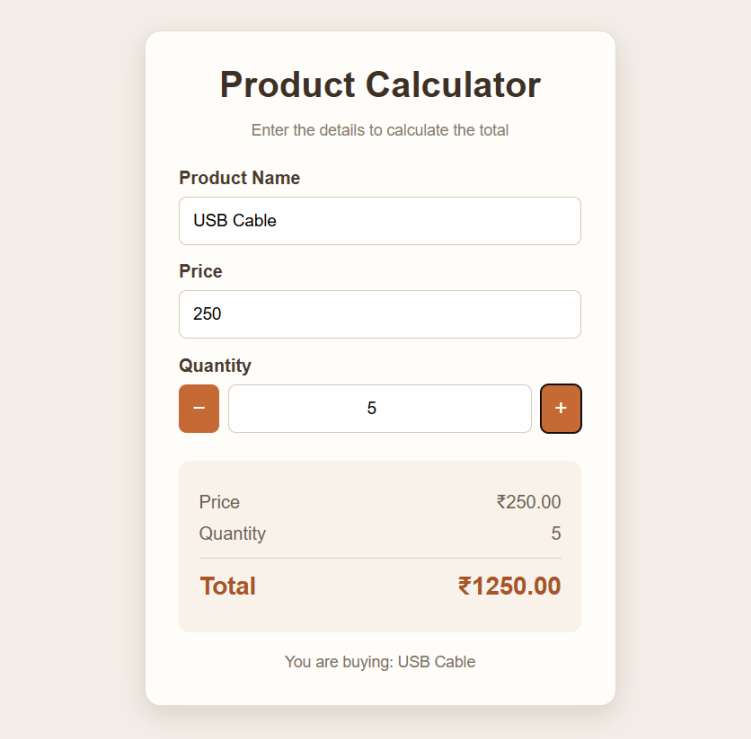
TC-02 Output:



TC_03 Output:

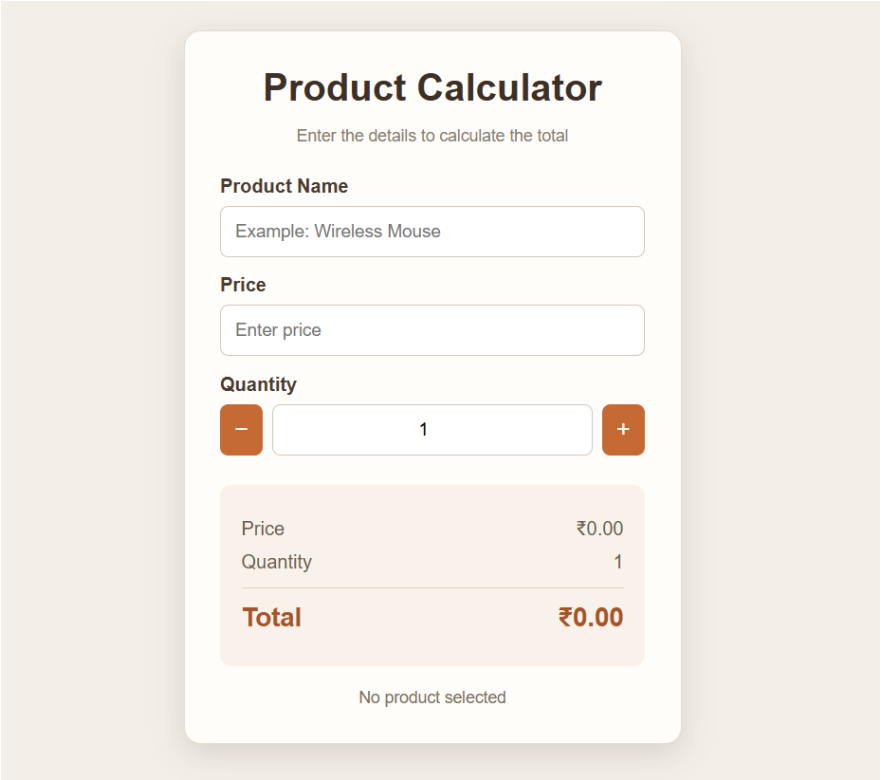


TC-04 Output:


Product Calculator
Enter the details to calculate the total
Product Name
USB Cable
Price
250
Quantity
−
5
+
Price
₹250.00
Quantity
5
Total
₹1250.00
You are buying: USB Cable

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9. OUTPUT:


Product Calculator
Enter the details to calculate the total
Product Name
Example: Wireless Mouse
Price
Enter price
Quantity
−
1
+
Price
₹0.00
Quantity
1
Total
₹0.00
No product selected

10. RESULT

Thus, a responsive **Product Calculator** was successfully developed using **HTML, CSS and JavaScript**. The application accepts product name, price and quantity from the user, calculates the total price dynamically, and updates the webpage using DOM manipulation and event listeners. Quantity increment and decrement operations, input validation, and dynamic product information display were also implemented successfully.